

Finerenone, eplerenone, and spironolactone in heart failure with preserved ejection fraction: a Bayesian network meta-analysis of efficacy and safety

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Abstract

Aims

The comparative effects of mineralocorticoid receptor antagonists (MRAs) in heart failure with preserved ejection fraction (HFpEF) or mildly reduced ejection fraction (HFmrEF) remain uncertain, including potential differences between steroidal MRAs and finerenone. This Bayesian network meta-analysis aimed to compare the efficacy and safety of finerenone, eplerenone, and spironolactone vs placebo in HFpEF/HFmrEF.

Methods

We searched Pubmed, Cochrane, and Embase for studies focused on MRA treatment in HFpEF and/or HFmrEF. Eight randomized controlled trials enrolling adults with HFpEF or HFmrEF (LVEF $\geq 40\%$) were analysed in a fixed-effects Bayesian network meta-analysis. Risk ratios (RRs) with 95% credible intervals (CrIs) were estimated for hospitalization for HF, cardiovascular death, and hyperkalaemia. Heterogeneity was assessed using I^2 .

Results

Across a network comprising 10 644 patients, mean age 70.2; 48% women; mean LVEF 54.9%, finerenone reduced hospitalization for HF vs placebo (RR 0.84; 95% CrI 0.75–0.93), whereas spironolactone (RR 0.86; 95% CrI 0.73–1.01) and eplerenone (RR 0.86; 95% CrI 0.59–1.26) showed non-significant effects. None of the MRAs achieved a statistically significant reduction in cardiovascular mortality vs placebo: finerenone (RR 0.93; 95% CrI 0.78–1.10), spironolactone (RR 0.93; 95% CrI 0.76–1.14), and eplerenone (RR 1.10; 95% CrI 0.29–5.69). All MRAs increased hyperkalaemia vs placebo: finerenone (RR 2.06; 95% CrI 1.77–2.41), spironolactone (RR 2.13; 95% CrI 1.81–2.53), and eplerenone (RR 2.17; 95% CrI 0.75–7.97).

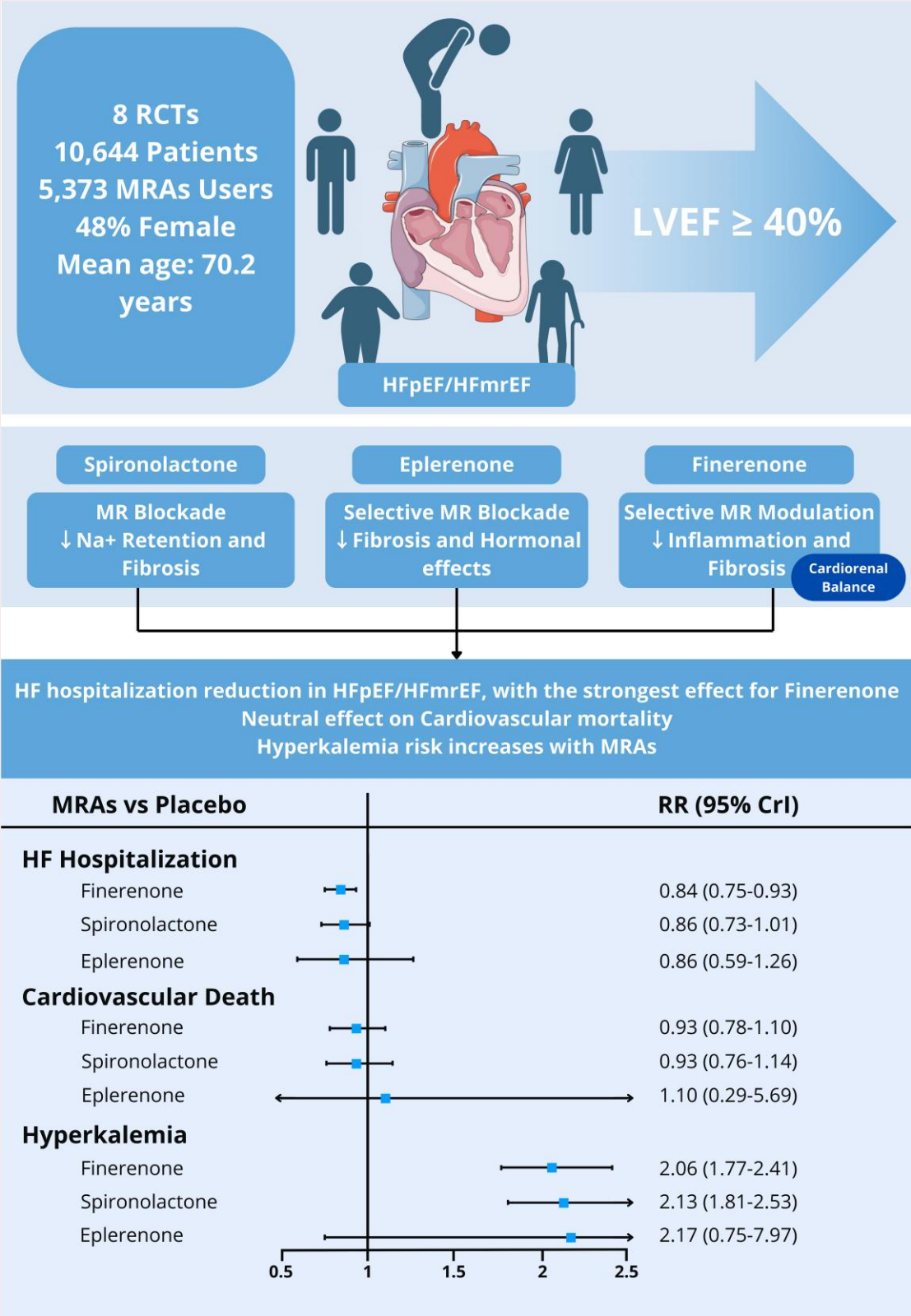
Conclusion

In HFpEF/HFmrEF, finerenone was the only MRA to significantly reduce hospitalization for HF, while no agent significantly reduced cardiovascular mortality and hyperkalaemia risk increased across therapies. Overall, finerenone may offer the most favourable efficacy–safety balance among MRAs, pending confirmation in larger dedicated trials.

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Graphical Abstract



Finerenone, eplerenone, and spironolactone in HFpEF/HFmrEF. Abbreviations: RCT, randomized controlled trial; MRA, mineralocorticoid receptor antagonist; LVEF, left ventricular ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFmrEF, heart failure with mildly reduced ejection fraction; MR, mineralocorticoid receptor; HF, heart failure; RR, risk ratio; CrI, credible interval.

Keywords

Heart failure with preserved ejection fraction • Heart failure with mildly reduced ejection fraction • Mineralocorticoid receptor antagonists • Finerenone • Spironolactone • Eplerenone

Introduction

Heart failure (HF) is a heterogeneous syndrome characterized by symptoms and signs resulting from structural or functional abnormalities that impair ventricular filling or ejection of blood.^{1,2} Diagnosis requires elevated natriuretic peptide levels or objective evidence of cardiogenic pulmonary or systemic congestion.^{3,4} The classification of HF by left ventricular ejection fraction (LVEF) is a key parameter for prognostic and therapeutic stratification. Thus, HF with reduced ejection fraction (HFrEF) is defined as LVEF $\leq 40\%$, HF with mildly reduced ejection fraction (HFmrEF) as LVEF between 41% and 49%, and HF with preserved ejection fraction (HFpEF) as LVEF $\geq 50\%$.^{1,2}

Heart failure with preserved ejection fraction represents 50% of all cases of HF,⁵ with mortality ranges between 15% at 1 year and 75% at 5–10 years after hospitalization.⁶ The number of patients with HFpEF continues to increase, driven by population ageing, demographic expansion, and the growing prevalence of comorbidities, such as obesity, hypertension, and diabetes.⁵ Unlike HFrEF, for which therapy with renin–angiotensin system inhibitors, beta-blockers, mineralocorticoid receptor antagonists (MRAs), and sodium–glucose cotransporter-2 inhibitors (SGLT2i) is well established and strongly endorse, therapeutic options for HFpEF and HFmrEF remain limited.^{1,2}

MRAs, namely the steroidal agents spironolactone and eplerenone, and the non-steroidal agent finerenone, act by blocking aldosterone action at the mineralocorticoid receptor, expressed in the kidneys, heart, and vasculature. Excess mineralocorticoid receptor activation promotes sodium retention, potassium loss, inflammation, myocardial fibrosis, and adverse cardiac remodelling, pathophysiological mechanisms that play a central role in HF.^{7,8} Through attenuation of these effects, MRAs limit neurohormonal overactivation, improve ventricular structure and function, and may reduce morbidity and mortality.^{7,8}

To date, no head-to-head comparisons among spironolactone, eplerenone, and finerenone have been conducted in patients with HFmrEF or HFpEF. This evidence gap limits the assessment of their relative efficacy and safety. Network meta-analysis (NMA) provides a robust framework for comparing multiple therapies by integrating both direct and indirect evidence.⁹ Therefore, we performed a systematic review and Bayesian NMA evaluating the efficacy and safety of finerenone, eplerenone, and spironolactone in patients with HFmrEF and HFpEF, aiming to provide comparative evidence that can inform clinical decision-making and future guideline recommendations.

Methods

This systematic review and network meta-analysis were performed and reported according to the Cochrane Collaboration Handbook for Systematic Reviews of Interventions⁹ and Preferred Reporting Items for Systematic Reviews and Meta-Analyses for Network Meta-Analyses.^{10,11} Protocol registered in the International Prospective Register of Systematic Reviews (PROSPERO) (registration number: CRD420251238519).

Search strategy and data extraction

We systematically searched on PubMed, Embase, and Cochrane databases from inception to September 2025, without any temporal restrictions imposed with the following search terms in all three databases: ('Heart failure with preserved ejection fraction' OR 'Heart failure with mildly reduced ejection fraction' OR 'HFpEF' OR 'HFmrEF') AND ('Spironolactone' OR

'Eplerenone' OR 'Finerenone'). The complete search strategy applied to each database is provided in [Supplementary Table S1](#).

Furthermore, the reference lists of included studies and relevant review articles were manually screened to identify additional eligible studies. Two authors (V.B. and H.S.S.) independently reviewed and extracted data according to predefined eligibility criteria. Disagreements were resolved by consensus with a third reviewer (S.C.L.).

Eligibility criteria

Eligible studies were those that met the following criteria: (1) randomized controlled trials (RCTs); (2) trials evaluating finerenone, spironolactone, or eplerenones; (3) studies including a placebo or standard-care control group; (4) enrolling adult patients (≥ 18 years) with HFpEF or HFmrEF, or with preserved LVEF and clinical characteristics consistent with these conditions, defined LVEF $\geq 40\%$; and (5) studies reporting at least one prespecified outcome of interest, including hospitalization for HF, cardiovascular death, or hyperkalaemia.

Because included trials applied heterogeneous LVEF entry thresholds (ranging from $\geq 40\%$ to $\geq 55\%$), the pooled population reflects a spectrum spanning HFmrEF and HFpEF rather than a uniform EF-defined phenotype. To improve interpretability and assess transferability across EF strata, trial-specific EF inclusion criteria were explicitly summarized and visualized ([Table 1](#)).

We excluded (1) non-comparative or cross-sectional studies, (2) overlapping datasets (only the study with the largest population or longest follow-up was included), (3) studies with no outcomes of interest, and (4) paediatric population. No restrictions were applied regarding publication date or language.

Endpoints

Outcomes included cardiovascular death and hospitalization for HF, defined as any unplanned hospital admission or urgent visit primarily due to clinical worsening of HF, evidenced by new or worsening symptoms and/or signs of congestion that required intensification of HF management. Hyperkalaemia, defined as a serum potassium ≥ 5.5 mmol/L, was the prespecified safety outcome.

Quality assessment

The risk of bias in the included RCTs was assessed using the Cochrane Risk of Bias 2 (RoB 2) tool.¹² Studies were rated as low risk, some concerns, or high risk of bias across five domains: selection, performance, detection, attrition, and reporting biases. Two reviewers (V.B. and H.S.S.) independently assessed risk of bias, and discrepancies were resolved by consensus with a third reviewer (C.d.O.F.B.).

Statistical analysis

A Bayesian NMA was conducted to estimate posterior mean effects, reported as RRs with 95% credible intervals (CrIs) for all outcomes, using fixed-effects consistency models. An estimated effect was considered statistically significant when the 95% CrI did not include unity. Network plots showed treatment connections and the relative contribution of each study to the overall evidence base. Models were computed using Markov Chain Monte Carlo simulations with four chains initialized with overdispersed starting values. Gibbs sampling was performed for a total of 100 000 iterations, with a burn-in phase of 20 000 iterations and a thinning interval of 50. Vague priors were used in all models. Model convergence and consistency were evaluated using the Brooks–Gelman–Rubin diagnostic, with convergence assessed visually through trace plots and numerically using the Potential Scale Reduction Factor (PSRF), where values < 1.05 indicated adequate model stability. Because the evidence network was star-shaped, with all active treatments compared only with placebo and no closed evidence loops, node-splitting for local inconsistency assessment was not applicable.

Table 1 Baseline characteristics of included studies

Study	Study design	MRA	LVEF inclusion criteria (%)	Dose	Population, N	Age, years; mean ± SD	Female sex, N (%)	LVEF (%); mean ± SD	BMI (kg/m ²); mean ± SD	Hypertension, N (%)	Diabetes mellitus, N (%)	Follow-up, months
Deswal 2011	RCT	Eplerenone	≥50%	50 mg/day	44 (21/23)	72.2 ± 9.8/68.7 ± 9.1	1 (4.8%)/2 (8.7%)	62.1 ± 5.0/62.5 ± 7.5	30.1 ± 6.1/34.6 ± 5.8	21 (100%)/23 (100%)	13 (61.9%)/14 (60.9%)	6.4
Edelmann 2013	RCT	Spirolactone	≥50%	25 mg/day	422 (213/209)	67.0 ± 8.0/67.0 ± 8.0	111 (52%)/110 (53%)	67.0 ± 8.0/68.0 ± 7.0	28.9 ± 3.6/28.9 ± 3.6	197 (92%)/190 (91%)	36 (17%)/34 (16%)	12
Kampourides 2012	RCT	Eplerenone	≥40%	25 mg/day	303 (201/102)	58.7 ± 13.3 ^a	54 (17.8%) ^a	56.7 ± 5.2 ^a	29.0 ± 4.4 ^a	174 (58%) ^a	60 (20%) ^a	24
Kosmala 2016	RCT	Spirolactone	>50%	25 mg/day	131 (64/67)	66.3 ± 7.7/67.6 ± 9.1	56 (88%)/54 (81%)	72.6 ± 9.0/71.4 ± 9.0	30.7 ± 4.5/29.7 ± 4.6	59 (92%)/61 (91%)	25 (39%)/27 (40%)	6
Kurrilmeyer 2014	RCT	Spirolactone	≥50%	25 mg/day	48 (24/24)	66.3 ± 2.2/64.4 ± 1.6	24 (100%)/24 (100%)	62.5 ± 1.2/62.9 ± 1.2	29.4 ± 2.2/26.3 ± 1.2	21 (87.5%)/19 (79.2%)	12 (50%)/6 (25%)	6
Pitt 2014	RCT	Spirolactone	≥45%	15–45 mg/day	3445 (1722/1723)	68.7 ± 11.4/68.3 ± 11.0	888 (51.6%)/887 (51.5%)	56.0 ± 7.4/56.3 ± 8.2	31.3 ± 6.7/31.3 ± 6.7	1567 (91%)/1580 (91.9%)	565 (32.8%)/553 (32.2%)	39.6
Shantsla 2020	RCT	Spirolactone	≥ 55%	25 mg/day	250 (125/125)	72.7 ± 6.7/72.3 ± 8.2	28 (22%)/31 (25%)	59.0 ± 3.7/59.0 ± 5.2	29.3 ± 5.2/30.0 ± 5.9	NA	24 (19%)/21 (17%)	24
Solomon 2024	RCT	Finerenone	≥40%	20–40 mg/day	6001 (3003/2998)	71.9 ± 9.6/72.0 ± 9.7	1355 (45.1%)/1377 (45.9%)	52.6 ± 7.8/52.5 ± 7.8	29.9 ± 6.1/30.0 ± 6.1	2640 (87.9%)/2685 (89.6%)	1217 (40.5%)/1222 (40.8%)	32

For patient characteristics, values are presented as MRA/control.

BMI, body mass index; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; N, number of participants; NA, not available; SD, standard deviation.

^aReported values are provided for the total population when group-specific data were not available.

Heterogeneity was assessed using the I², with values <25% indicating low heterogeneity. Both fixed- and random-effects models were generated, and model fit was evaluated using the deviance information criterion (DIC). The model with the lowest DIC was considered to provide the best fit; however, when the difference in DIC between the two models was <3, the fixed-effects was preferred. Accordingly, a Bayesian fixed-effects model was selected based on the low heterogeneity and comparable DIC performance. Ranking probabilities for treatment efficacy and safety were estimated to establish a comparative hierarchy among interventions. These rankings were summarized as Surface Under the Cumulative Ranking Curve (SUCRA) values and graphically illustrated using rankograms. All analyses were conducted using R software version 4.5.1 (R Foundation for Statistical Computing, Vienna, Austria) equipped with the ‘gemtc’ package.

Results

Study selection and characteristics

The study selection process is summarized in [Supplementary Figure S1](#). Our systematic search identified 2544 potential records. After removing duplicates and ineligible studies, 47 articles underwent full-text review according to our predefined inclusion criteria. Following screening, eight RCTs^{13–20} comprising a total of 10 644 patients were deemed eligible and included in the final analysis.

The baseline characteristics of the included studies are summarized in [Table 1](#). In the pooled population, the mean age was 70.2 years, 48% of patients were women, the mean body mass index was 30.3 kg/m², and the mean LVEF was 54.9%, with all estimates weighted by sample size. Across all trials, the duration of follow-up ranged from 6 to 40 months, with a weighted mean follow-up of 32.7 months. Given the substantial variation in trial size and follow-up, particularly among eplerenone studies, event counts for hospitalization and cardiovascular death are explicitly reported for each trial ([Supplementary Table S2](#)) to contextualize the precision of network estimates. Notably, EF inclusion criteria varied substantially across trials, with lower thresholds ranging from ≥40% to ≥55%, potentially overlapping HFmrEF and HFpEF populations. This heterogeneity is visually summarized in [Table 1](#), highlighting the spectrum of EF definitions informing the evidence network.

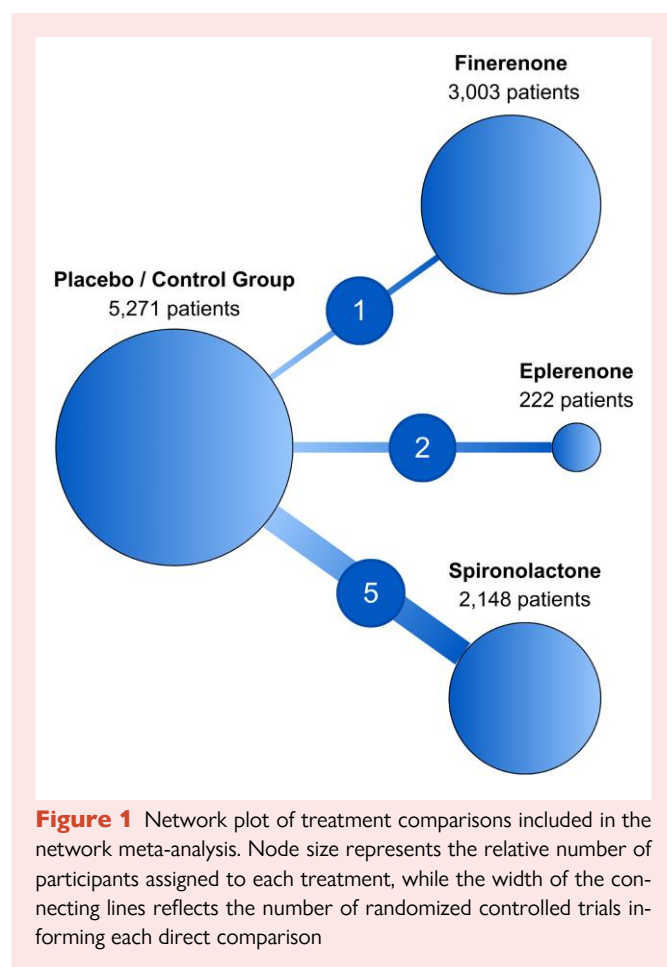
Structure of network

[Figure 1](#) shows the network of interventions, consisting of four treatment arms: placebo, spironolactone, eplerenone, and finerenone. All active therapies were evaluated against placebo, which served as the common comparator. Five RCTs investigated spironolactone vs placebo, two assessed eplerenone vs placebo (one including a control group without eplerenone), and one trial evaluated finerenone vs placebo. No direct head-to-head comparisons among the active interventions were available. Consequently, the network displayed a star-shaped configuration, with placebo forming the central node connecting all active treatments and no closed loops between interventions.

Efficacy endpoints

Of the eight included trials, the primary efficacy outcome of hospitalization for HF was pooled across 10 644 patients, comparing finerenone, spironolactone, eplerenone, and placebo. The Bayesian NMA demonstrated that finerenone was significantly more effective in reducing hospitalizations for HF compared with placebo (RR 0.84; 95% CrI, 0.75–0.93). Spironolactone showed a non-significant trend towards benefit (RR 0.86; 95% CrI, 0.73–1.01), whereas eplerenone demonstrated a similar, non-significant effect (RR 0.86; 95% CrI, 0.59–1.26).

Indirect head-to-head comparisons among the MRAs showed no statistically significant differences. When compared with spironolactone, finerenone (RR 0.97; 95% CrI, 0.80–1.19) and eplerenone (RR 1.00; 95% CrI, 0.66–1.53) exhibited comparable efficacy, while placebo



was associated with a higher relative risk of hospitalization for HF (RR 1.16; 95% CrI, 0.99–1.37). These results for the primary efficacy endpoint are summarized in [Table 2](#).

In terms of efficacy ranking, finerenone appeared to be the most effective treatment in reducing hospitalizations for HF, with a SUCRA value of 0.72, followed by spironolactone (0.62), eplerenone (0.58), and placebo (0.08). No significant inconsistency or heterogeneity was observed across the network ($I^2 = 0\%$). All Bayesian models demonstrated adequate convergence, with PSRF values <1.05 for all parameters and outcomes. Analyses were performed using both fixed- and random-effects Bayesian models, which yielded consistent estimates across all endpoints, supporting the robustness of the results ([Figure 2](#)).

One trial¹⁴ did not report cardiovascular death, leaving a total of seven trials encompassing 10 222 patients that reported data for this secondary efficacy endpoint ([Supplementary Figure S2](#)). In the Bayesian NMA, none of the active MRAs achieved a statistically significant reduction in cardiovascular mortality vs placebo. Finerenone showed a modest, non-significant decrease in risk (RR 0.93; 95% CrI, 0.78–1.10), spironolactone demonstrated a comparable estimate (RR 0.93; 95% CrI, 0.76–1.14), and eplerenone presented a wide and imprecise interval (RR 1.10; 95% CrI, 0.29–5.69), reflecting limited evidence.

When comparing the MRAs indirectly, the results remained consistent across pairwise contrasts. Finerenone and spironolactone had similar efficacy (RR 1.00; 95% CrI, 0.77–1.30), and eplerenone did not differ significantly from either comparator (RR 1.18; 95% CrI, 0.31–6.14 vs spironolactone). None of the indirect comparisons suggested superiority of one agent over another. The corresponding estimates for cardiovascular mortality are presented in [Table 2](#).

Based on ranking probabilities, finerenone had the highest likelihood of being the most effective therapy for reducing cardiovascular death (SUCRA = 0.64), followed by spironolactone (0.61), eplerenone (0.42), and placebo (0.33). The analysis revealed no meaningful inconsistency or heterogeneity across the network ($I^2 = 0\%$) ([Figure 3](#)).

Safety endpoint

Eight trials including a total of 10 629 patients were analysed for the safety outcome of hyperkalaemia ([Supplementary Figure S3](#)). The Bayesian NMA demonstrated that all MRAs were associated with an increased risk of hyperkalaemia compared with placebo. Finerenone (RR 2.06; 95% CrI, 1.77–2.41), spironolactone (RR 2.13; 95% CrI, 1.81–2.53), and eplerenone (RR 2.17; 95% CrI, 0.75–7.97) all showed higher event rates than placebo.

Indirect comparisons among the MRAs revealed no statistically significant differences. When compared with spironolactone, finerenone (RR 0.97; 95% CrI, 0.77–1.21), and eplerenone (RR 1.02; 95% CrI, 0.34–3.78) showed comparable risks of hyperkalaemia, while placebo was associated with a markedly lower risk (RR 0.47; 95% CrI, 0.40–0.55). Comparable results were observed when finerenone or eplerenone served as reference treatments. Although not statistically significant, finerenone demonstrated a numerically lower incidence of hyperkalaemia compared with spironolactone, suggesting a potentially more favourable safety profile. These estimates for hyperkalaemia risk are summarized in [Table 2](#).

According to the SUCRA, among the active treatments, finerenone ranked as the safest agent (SUCRA = 0.38), followed by eplerenone (0.34) and spironolactone (0.30). As expected, placebo had the highest safety ranking (SUCRA = 0.97). No relevant inconsistency or heterogeneity was detected across the network ($I^2 = 0\%$) ([Figure 4](#), [Table 2](#)).

Quality assessment

The risk of bias across the included trials was judged to be low to moderate ([Supplementary Figure S4](#)). Most studies were at low risk of bias across key domains, including randomization, adherence to intended interventions, and outcome measurement. One trial (19) was classified as having high risk of bias due to its open-label design, which introduced potential bias in outcome measurement. The consistency of results across fixed- and random-effects models suggest this is unlikely to affect the overall conclusions ([Supplementary Table S3](#)).

Discussion

In this Bayesian NMA of eight RCTs involving patients with HFpEF or HFmrEF, none of the evaluated MRAs significantly reduced cardiovascular mortality compared with placebo; however, CrIs were wide, particularly for eplerenone, reflecting limited event rates, small trial sizes, and reduced statistical power for mortality inference. In addition, follow-up duration in the finerenone trial was shorter than in several spironolactone studies, which may further constrain the detection of long-term mortality effects. However, finerenone demonstrated a statistically significant reduction in hospitalizations for HF and a favourable balance between efficacy and safety. At the same time, spironolactone showed a non-significant trend towards hospitalization reduction, and eplerenone exhibited largely neutral effects. Hyperkalaemia incidence increased across all MRAs, with comparable relative risk between the steroidal and non-steroidal classes, but with a lower overall severe event burden with finerenone.

The clinical implications of these findings are particularly relevant given the growing economic burden associated with HFpEF in the United States. HF-related hospitalizations account for a significant proportion of direct costs, with HFpEF admissions averaging US\$10 000–\$30 000 per episode, depending on comorbidity burden. Annual HF

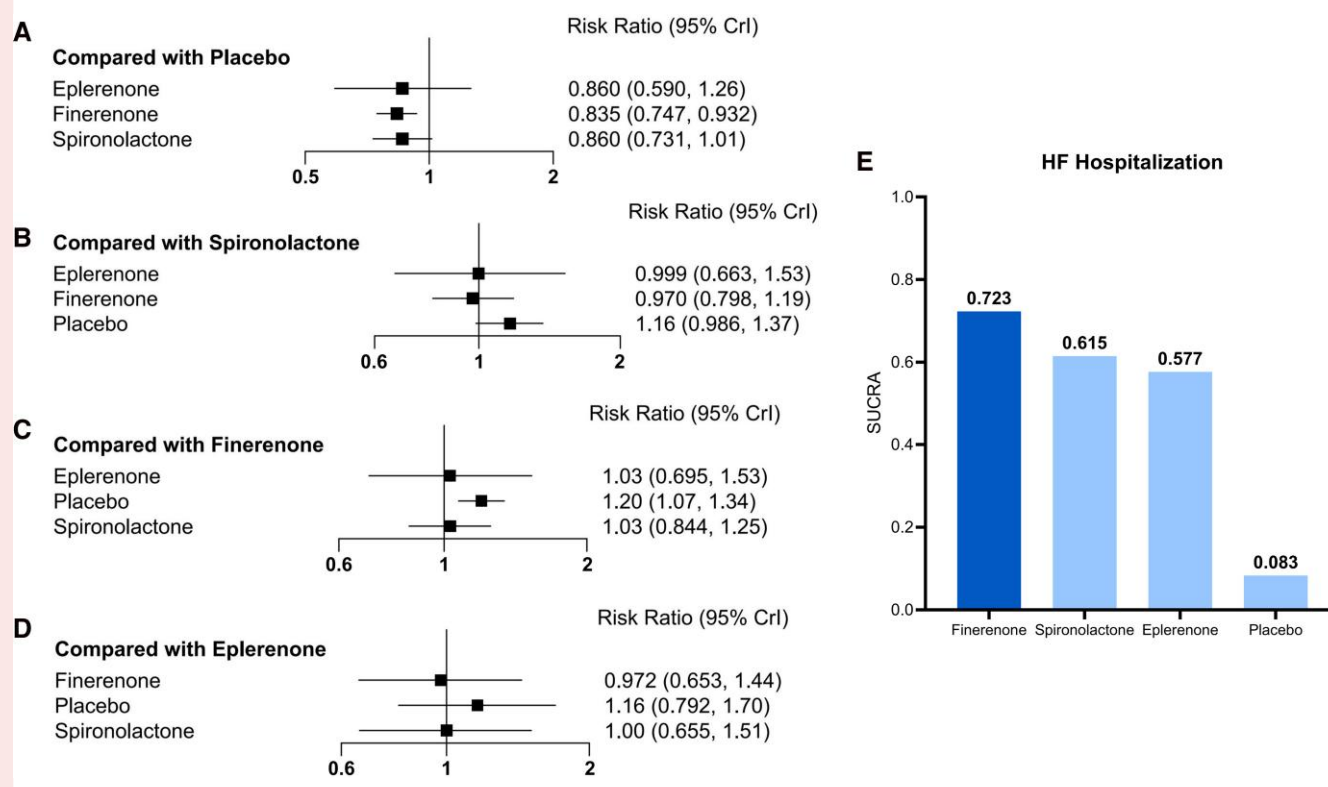


Figure 2 Forest plots and surface under the cumulative ranking curve values for hospitalization for heart failure. (A–D) Show forest plots from the network meta-analysis comparing each treatment against the reference treatment (bold font), which corresponds to (A) placebo; (B) spironolactone; (C) finerenone; and (D) eplerenone. A risk ratio < 1 indicates a lower risk of hospitalizations for heart failure with treatment 1 compared with treatment 2. (E) Shows the surface under the cumulative ranking curve values for hospitalizations for heart failure. Surface under the cumulative ranking curve values represent the probability that each treatment is the most effective for reducing hospitalizations for heart failure. Higher surface under the cumulative ranking curve percentages indicate a higher likelihood of superior performance among the compared mineralocorticoid receptor antagonists. 95% CrI, 95% credible interval

expenditures in the U.S. exceed US\$30 billion, with projections that will surpass US\$70 billion by 2030, driven mainly by recurrent hospitalizations and an ageing population.^{21,22} Because HFpEF patients experience frequent decompensations despite preserved systolic function, therapies capable of reducing hospitalization risk, such as finerenone in this study, are economically meaningful and may help alleviate the disproportionate healthcare burden in this phenotype.²²

Mechanistically, MRAs counteract aldosterone-induced sodium retention, inflammation, microvascular dysfunction, and myocardial fibrosis.^{7,8} Spironolactone and eplerenone are steroidal MRAs that bind mineralocorticoid and, to varying degrees, androgen and progesterone receptors, which can contribute to hormonal side effects such as gynaecomastia and menstrual irregularities.⁷ Finerenone is a selective non-steroidal MRA with reduced tissue accumulation and more potent anti-inflammatory and anti-fibrotic gene modulation, potentially yielding a better cardiorenal profile and reduced hyperkalaemia severity.^{7,8} The comparative pharmacologic characteristics are summarized in Table 3.

Our findings align with and extend evidence from contemporary HFpEF literature. The TOPCAT trial evaluating spironolactone in HFpEF demonstrated regional variability and modest reduction in hospitalization for HF without mortality benefit.¹⁸ EMPHASIS-HF and RALES firmly established benefit in HFrEF, supporting a class effect in systolic dysfunction but highlighting uncertainty in preserved EF phenotypes.^{23,24} Finerenone's benefit in HFpEF was most recently supported

by FINEARTS-HF, showing a significant reduction in HF events and a favourable safety profile.²⁰ These evolving data, together with our results, suggest that while steroidal MRAs may exert beneficial remodelling effects, non-steroidal MRAs such as finerenone may provide incremental benefit in HFpEF due to enhanced molecular targeting and improved tolerability.^{3,25} Future real-world comparative studies and head-to-head RCTs are needed to confirm these relative advantages.²⁶

These mechanistic and pharmacologic distinctions raise the question of whether all MRAs confer similar benefit in HFmrEF/HFpEF. A recent individual patient-level meta-analysis by Jhund *et al.*²⁶ provided an essential benchmark by demonstrating that MRAs reduce composite cardiovascular events across the ejection fraction spectrum, with greater benefit observed in HFrEF and attenuated effects in HFmrEF/HFpEF (particularly for mortality). Our study advances this evidence base by specifically isolating the HFmrEF/HFpEF population and, crucially, by comparing individual agents rather than the MRA class as a whole. Through indirect head-to-head Bayesian comparisons, we highlight clinically meaningful differences in efficacy and safety among spironolactone, eplerenone, and finerenone, providing therapeutic granularity that was not possible in prior analyses. This approach addresses a key evidence gap by helping clinicians not only determine whether an MRA should be considered in HFmrEF/HFpEF, but also which MRA may be most appropriate based on the patient's clinical profile and risk-benefit trade-offs. Taken together, these findings refine therapeutic

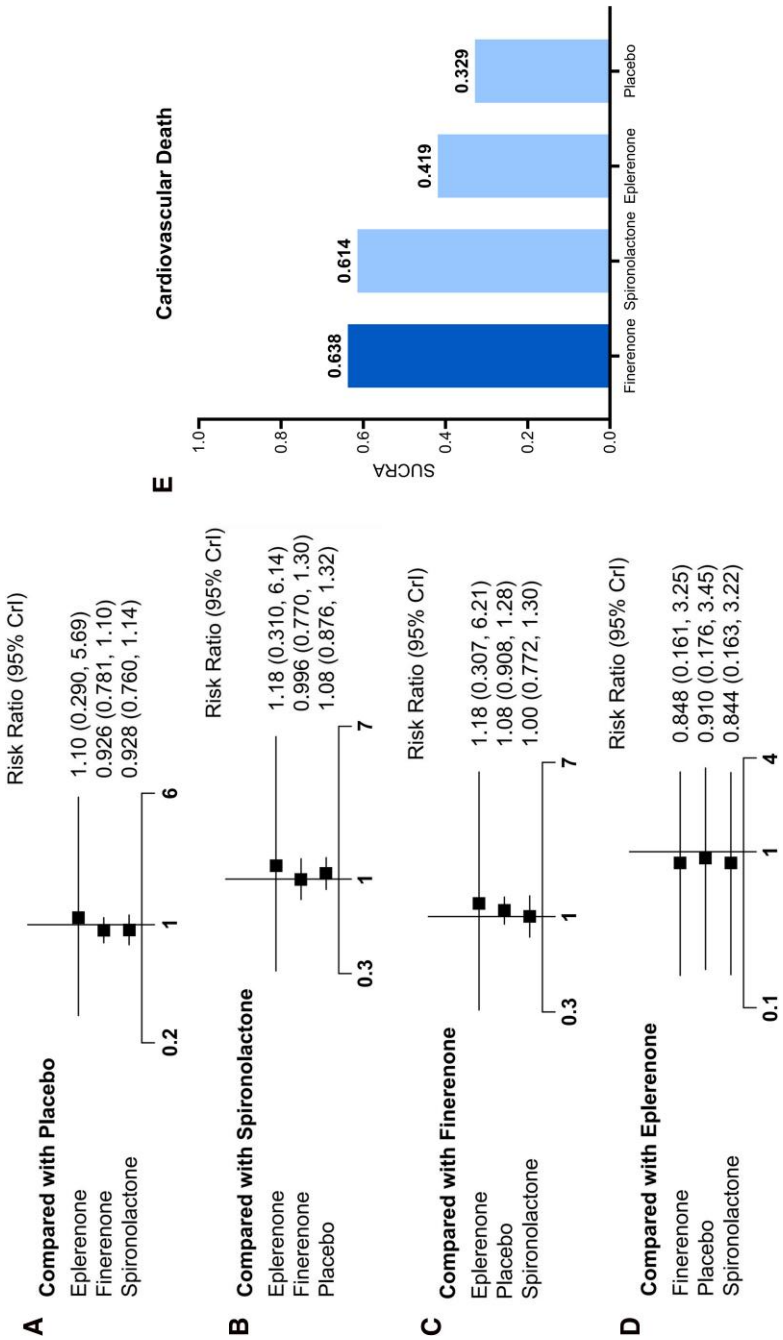


Figure 3 Forest plots and surface under the cumulative ranking curve values for cardiovascular death. (A–D) show forest plots from the network meta-analysis comparing each treatment against the reference treatment (bold font), which corresponds to (A), placebo; (B) spironolactone; (C) finerenone; and (D) eplerenone. A risk ratio (RR) <1 indicates a lower risk of cardiovascular death with treatment 1 compared with treatment 2. (E) Shows the surface under the cumulative ranking curve values for cardiovascular death. Surface under the cumulative ranking curve values represent the probability of each treatment being the most effective option for reducing cardiovascular death. Higher surface under the cumulative ranking curve percentages indicate a higher likelihood of superior performance among the compared mineralocorticoid receptor antagonists. 95% CrI, 95% credible interval

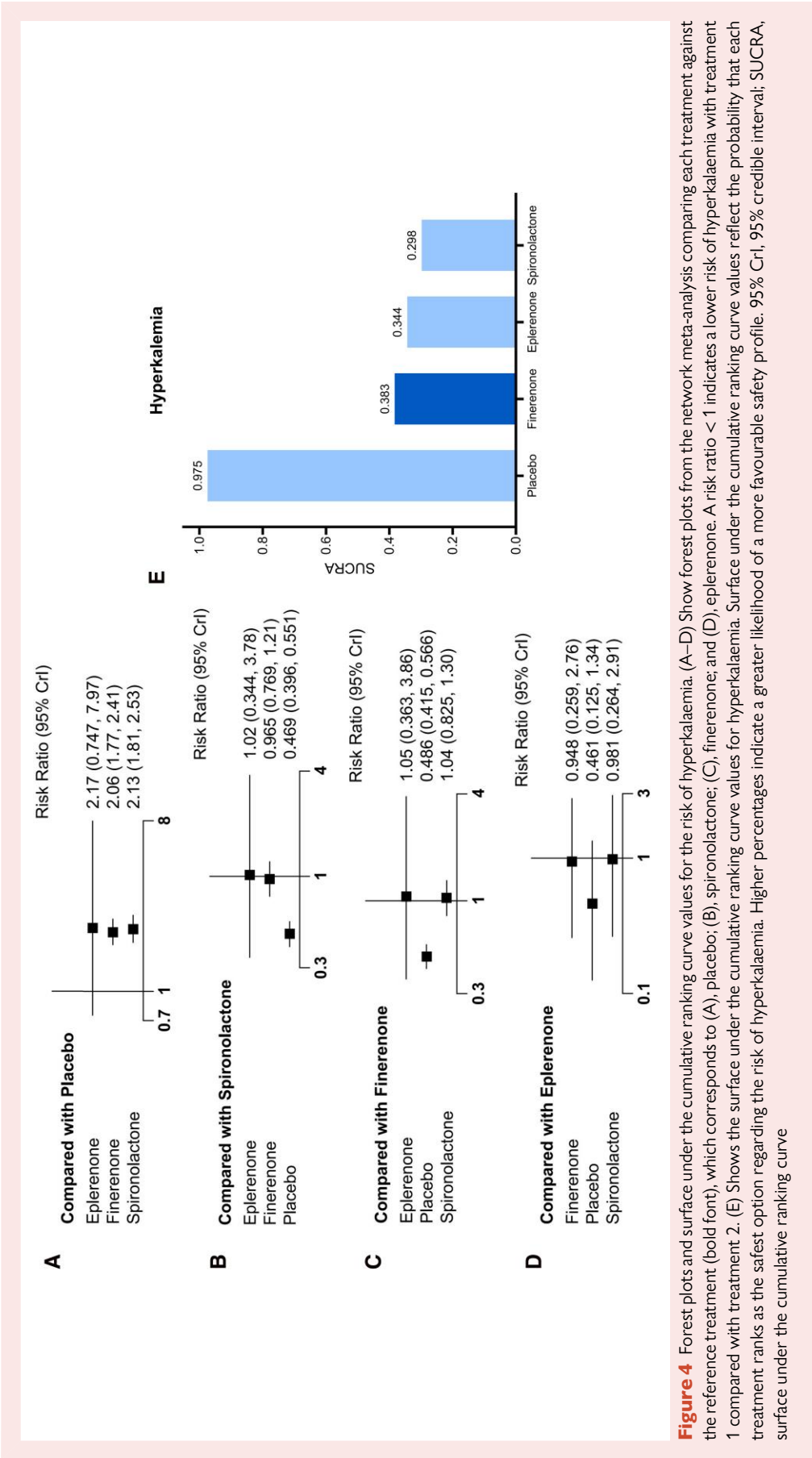


Figure 4 Forest plots and surface under the cumulative ranking curve values for the risk of hyperkalaemia. (A–D) Show forest plots from the network meta-analysis comparing each treatment against the reference treatment (bold font), which corresponds to (A), placebo; (B), spironolactone; (C), finerenone; and (D), eplerenone. A risk ratio < 1 indicates a lower risk of hyperkalaemia with treatment 1 compared with treatment 2. (E) Shows the surface under the cumulative ranking curve values for hyperkalaemia. Surface under the cumulative ranking curve values reflect the probability that each treatment ranks as the safest option regarding the risk of hyperkalaemia. Higher percentages indicate a greater likelihood of a more favourable safety profile. 95% CrI, 95% credible interval; SUCRA, surface under the cumulative ranking curve

Table 2 League table of network meta-analysis results for hospitalization for heart failure, cardiovascular death, and hyperkalaemia

	Eplerenone	Finerenone	Placebo	Spironolactone
HF hospitalization - RR (95% CrI)				
Eplerenone	—	0.972 (0.653, 1.440)	1.163 (0.792, 1.696)	1.001 (0.655, 1.509)
Finerenone	1.029 (0.695, 1.530)	—	1.198 (1.073, 1.339)	1.030 (0.844, 1.253)
Placebo	0.860 (0.590, 1.262)	0.835 (0.747, 0.932)	—	0.860 (0.731, 1.014)
Spironolactone	0.999 (0.663, 1.526)	0.970 (0.798, 1.185)	1.162 (0.986, 1.369)	—
Cardiovascular death - RR (95% CrI)				
Eplerenone	—	0.848 (0.161, 3.253)	0.910 (0.176, 3.449)	0.844 (0.163, 3.224)
Finerenone	1.180 (0.307, 6.207)	—	1.080 (0.908, 1.280)	1.004 (0.772, 1.299)
Placebo	1.098 (0.290, 5.692)	0.926 (0.781, 1.101)	—	0.928 (0.760, 1.142)
Spironolactone	1.185 (0.310, 6.143)	0.996 (0.770, 1.296)	1.077 (0.876, 1.316)	—
Hyperkalaemia - RR (95% CrI)				
Eplerenone	—	0.948 (0.259, 2.759)	0.461 (0.125, 1.339)	0.981 (0.264, 2.908)
Finerenone	1.055 (0.363, 3.865)	—	0.486 (0.415, 0.566)	1.036 (0.825, 1.301)
Placebo	2.169 (0.747, 7.969)	2.059 (1.766, 2.411)	—	2.133 (1.815, 2.527)
Spironolactone	1.019 (0.344, 3.785)	0.965 (0.769, 1.213)	0.469 (0.396, 0.551)	—

Each cell presents the risk ratio (RR) and corresponding 95% credible interval (CrI) comparing the column treatment against the row treatment. Values <1 favour the column treatment, indicating a lower risk of the outcome compared with the row treatment. Diagonal cells (—) indicate comparisons of a treatment with itself.

Table 3 Key features of mineralocorticoid receptor antagonists used in heart failure with preserved ejection fraction and heart failure with mildly reduced ejection fraction

Key features of MRAs used in HFpEF and HFmrEF			
Drug	Spironolactone	Eplerenone	Finerenone
Class	Steroidal MRA	Steroidal selective MRA	Non-steroidal MRA
Mechanism	Competitive MR blockade, partial sex-hormone receptor affinity	Selective MR blockade, minimal sex-hormone effects	Selective MR modulation, anti-fibrotic and anti-inflammatory signalling
Indications	HFpEF; may be considered in HFpEF	HFpEF, post-MI	CKD + T2DM; emerging HFpEF evidence
Contraindications	Hyperkalaemia, eGFR <30	Hyperkalaemia, eGFR <30	Hyperkalaemia, low GFR thresholds
Key adverse events	Hyperkalaemia, gynaecomastia, renal dysfunction, hypotension	Hyperkalaemia, renal dysfunction	Hyperkalaemia (lower severity), hypotension
Dose	25–50 mg/day	25–50 mg/day	20–40 mg/day
Approx. cost (US)	Low (generic)	Moderate	Higher (brand-only)

CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; GFR, glomerular filtration rate; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFpEF, heart failure with reduced ejection fraction; MI, myocardial infarction; MR, mineralocorticoid receptor; MRA, mineralocorticoid receptor antagonist; T2DM, type 2 diabetes mellitus.

decision-making in a population historically characterized by neutral trial results and limited targeted therapies.

HFpEF remains a complex syndrome with heterogeneous pathophysiology driven by systemic inflammation, microvascular dysfunction, metabolic risk factors, and comorbidity overlap.⁵ Despite therapeutic advances in SGLT2 inhibitors and emerging benefits of MRAs, unmet need persists. Innovation in HFpEF treatment requires precision phenotyping, mechanistic drug development, and integration of metabolic and anti-fibrotic pathways.²⁷ Future progress will likely include combinatorial neurohormonal therapy, inflammation-targeted pharmacology, integration of AI-guided phenomapping, and personalized treatment algorithms.²⁸ Broader adoption of digital monitoring and structured multidisciplinary care pathways will further support early detection,

treatment adherence, and hospitalization avoidance, core objectives in a syndrome currently more disabling than fatal.^{29,30}

Limitations

This NMA has several significant limitations that should be acknowledged. First, although we included only RCTs, the total number of eligible studies evaluating MRAs in HFpEF/HFmrEF remains limited, particularly when stratified by individual drug. Evidence for eplerenone and finerenone, while emerging, is still based on fewer trials than for spironolactone, which may reduce statistical power and increase uncertainty in indirect comparisons. Second, our analysis relied on study-level rather than individual patient-level data, the ability to adjust for patient-

specific characteristics such as comorbidity burden, diuretic use, natriuretic peptide levels, background guideline-directed therapy, and renal function, factors known to influence treatment response and hyperkalaemia risk in HFpEF.

Third, heterogeneity in HFpEF definitions, LVEF thresholds, regional enrolment patterns, and trial inclusion criteria may introduce clinical heterogeneity and limit generalizability. Fourth, because NMA enables indirect treatment comparisons, findings depend on the assumption of transitivity across study populations; unmeasured confounders may affect the validity of relative ranking among agents. In particular, evidence for eplerenone is derived from small, short-duration studies that were not primarily designed to evaluate hospitalization or mortality endpoints, increasing uncertainty and susceptibility to small-study effects.

Fifth, definitions and monitoring strategies for hyperkalaemia varied across trials, potentially influencing safety estimates. Although hypokalaemia has emerged as an increasingly recognized electrolyte complication in HF populations, particularly in diuretic-treated patients, it was inconsistently reported across the included RCTs and therefore could not be incorporated as a comparative safety endpoint in this network. Future trials should evaluate both hyperkalaemia and hypokalaemia to better characterize electrolyte risk profiles of MRAs in HFpEF.

Finally, the follow-up duration differed across included studies, and longer-term outcome data, particularly for finerenone, remain limited. As the therapeutic landscape evolves, future head-to-head RCTs and individual-patient pooled analyses will be essential to validate our findings and further refine drug-specific recommendations in HFpEF and HFmrEF.

Conclusion

Among patients with HFpEF and HFmrEF, none of the MRAs demonstrated a significant reduction in cardiovascular death; however, finerenone showed the most favourable balance of efficacy and safety, significantly reducing hospitalizations for HF with a tolerable hyperkalaemia profile. While spironolactone showed directional benefit and eplerenone remained neutral, these findings emphasize important pharmacologic distinctions within the MRA class and highlight the therapeutic promise of non-steroidal MR modulation in HFpEF. As HFpEF continues to rise in prevalence and complexity, individualized MRA selection may represent a key strategy for optimizing outcomes and warrants further validation in dedicated head-to-head trials.

Supplementary data

Supplementary data are available at [European Journal of Heart Failure](#) online.

Declarations

Disclosure of Interest

All authors report no relationships that could be construed as a conflict of interest. All authors take responsibility for all aspects of the reliability and freedom from bias of the data presented and their discussed interpretation and the corresponding author had final responsibility for the decision to submit for publication.

Data Availability

The data underlying this article are available in the article and in its online [supplementary material](#). Additional data are available on reasonable request to the corresponding author.

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Ethical Approval

Ethical Approval was not required.

Pre-registered Clinical Trial Number

None supplied.

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